# Literal diagram and construction joins for the $\mathrm{PSL}_3(4)$ catalogue

Finite source-recovery component

The fixed source is Gottschalk and Leemans, *Geometries for the group* PSL(3, 4), Halle Reports on Algebra and Geometry 99–30, the complete 22-page institutional preprint. Its PostScript and derived PDF are retained exactly. The 25-page journal edition, European Journal of Combinatorics 24 (2003), 267–291, has not been collated here. Statements about the printed diagrams below refer only to the preprint.

**Acquired inputs.** The accepted broad-primitive enumeration exhausts all 95 subgroup anchors in a fixed permutation group $G$ of order 20160, using the retained complete subgroup and maximal-inclusion inventories. Its full-automorphism fusion has 29 representatives of ranks 2–5. The separately accepted palette calculation, together with its exact serialization repair, provides complete right-coset partitions, incidence rows and actual $G$-actions for these representatives. The full-automorphism and GRAPE/nauty completeness premises remain explicit acquired inputs. They are not proved by checking positive permutation maps. The previously accepted projective construction component provides two rank-five models and eight literal truncation routes.

**Theorem 1.** *The retained literal calculations establish the following statements.*

1. *The numerical node data, all rank-two residues, local action kernels and local* 2*-transitivity flags are determined for each of the 29 acquired representatives. Within this complete catalogue, every preprint diagram has one representative matching its recorded invariants, subject to the three discrepancies in Lemma 1.*
2. *All 52 printed circular, Petersen and affine-plane bond labels have explicit canonical incidence maps, including their displayed polarity.*
3. *The accepted constructions labelled 5.1 and 5.2 are explicitly isomorphic to catalogue representatives 26 and 28, respectively. The eight acquired truncation routes also have complete object and incidence maps.*

*This is not an assertion that diagrams determine geometries in general. Abstract names of Borel groups, parabolics, kernels, automorphism groups and correlation groups require their own structural identifications.*

*Proof.* Choose the identity-coset chamber. For a set $J$ of remaining types, intersect the neighbour sets of all complementary chamber objects. These are exactly the objects of the base residue of type $J$. Acquired flag transitivity transports this calculation to every flag of the same type. A panel order is its cardinality minus one. In its actual stabilizer, the stabilizer of one panel object is tested for transitivity on the other objects. No local 2-transitivity assumption is imposed.

For every pair of types, all incidence and nonincidence rows of the resulting bipartite graph are retained. Breadth-first distances give eccentricities and the minimum cycle length; half that length is the gonality. A residue is a digon precisely when it is complete bipartite. At each base object, every element of its parabolic is tested on every object of its residue. The elements fixing all these objects are exactly the local action kernel. Only its order, not its abstract structure, is used in the numerical join.

For a circular bond, the degree-two objects map bijectively to all unordered pairs of the opposite object set. For a Petersen bond, an explicit map sends the ten points to the two-subsets of a five-set and the fifteen lines to disjoint pairs of such subsets. Every incidence row is checked. For an affine bond, sixteen points and twenty four-point lines have a unique line through each point pair and five parallel classes. Adding one point for each parallel class and a line at infinity gives a projective completion. An explicit 21-point bijection takes all its lines to the coordinate plane over $\mathbb{F}_4$. The recorded point/line maps also verify the source polarity.

Finally, the transcription records actual node positions, not presumed type numbers. All permutations compatible with the printed type sizes are checked against the complete residue data. The resulting table below records the unique catalogue index in each case; symmetric node permutations may remain. These tests compare the complete finite catalogue with the source, rather than using source counts as acceptance criteria. □

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|---|---|---|---|---|---|---|---|---|---|---|---|
| Source rank 2 | 2.1 | 2.2 | 2.3 | 2.4 | 2.5 | 2.6 | 2.7 | | | | |
| Catalogue | 1 | 4 | 2 | 5 | 3 | 6 | 7 | | | | |
| Source rank 3 | 3.1 | 3.2 | 3.3 | 3.4 | 3.5 | 3.6 | 3.7 | | | | |
| Catalogue | 9 | 13 | 10 | 11 | 14 | 12 | 8 | | | | |
| Source rank 4 | 4.1 | 4.2 | 4.3 | 4.4 | 4.5 | 4.6 | 4.7 | 4.8 | 4.9 | 4.10 | 4.11 |
| Catalogue | 15 | 24 | 19 | 22 | 23 | 18 | 17 | 16 | 21 | 20 | 25 |
| Source rank 5 | 5.1 | 5.2 | 5.3 | 5.4 | | | | | | | |
| Catalogue | 26 | 28 | 29 | 27 | | | | | | | |

**Construction maps and action conventions.** The projective component and census use different generator lists. Their complete generated permutation groups are verified equal on the same 21 labels. The accepted coordinate-to-retained map is reused. If necessary, an explicit duality is applied: a normal vector labels its perpendicular projective line, and the induced action on these 21 lines is transported back to the fixed point labels. This is checked to permute all 20160 group elements bijectively and preserve every edge of the generator Cayley graph, so it is an automorphism $\alpha$ of the actual group.

For every proposed object $x_i$, the generators of the actual parabolic $H_i$ fix $x_i$ after applying $\alpha$. Consequently

$$H_i g \longmapsto x_i^{\alpha(g)}$$

is well defined. The certificate additionally checks every coset member, the complete object bijections, all incidence and nonincidence rows, and the explicit type permutation. Separate checks of the saved maps verify 3,185 incidence rows and 6,370 actual generator-action rows.

**Lemma 1.** *In preprint diagrams 5.1 and 5.2, the printed order* 2 *at the unique 210-object node must be* 1 *for the stated constructions. In diagram 3.5, the missing bond between the two 120-object nodes must have gonality* 3 *if the other printed data and the stated geometry hypotheses hold.*

*Proof.* The literal construction maps send the 210-object node to the unordered point-pair type. Its panel has two objects and therefore order 1, as already verified in the accepted construction. The order-2 line node has 21 objects and cannot be confused with this node.

For 3.5 the 1120-object stabilizer has order $20160/1120 = 18$, while the printed Borel has order 3. Its rank-two residue thus has six chambers. Both remaining panel orders are 1, so both vertex degrees are two and each type has three objects. Residual connectedness forces a six-cycle, not a digon. The actual catalogue residue is retained as this literal six-cycle. This argument uses the source's own flag-transitive geometry hypotheses and does not assume that a diagram signature identifies a construction. □

**Scope and reproducibility.** The scripts retain all 29 literal residue tables, 52 canonical bond maps and ten construction/truncation maps. Bounded controls include 25 known graph distance/girth examples, six complete-graph incidence examples, damaged maps and missing endpoints. Two failed preliminary joins are preserved: one incorrectly required identical generator lists, and one omitted the needed duality. Neither negative attempt is a premise. No graph-isomorphism target or subgroup census is repeated by this component.

The source has no local-2-transitivity marker at 3.5 or 4.11, consistent with the actual panels. Earlier provisional transcription mistakes at those two rows are disclosed and corrected. The three discrepancies above do not establish an error in the uncollated journal edition. Remaining constructions, the general doubling hypotheses, abstract group-name claims and whole-source completion are outside this component. There is no new question, Lean, measurement or novelty claim.